

SMART INDUSTRIAL MACHINE HEALTH MONITORING AND PREDICTIVE SYSTEM

INPUT PORTS:

1.DC BARREL JACK:

DESCRIPTION:DC Power Jack Connector WR-DC, Male, Right Angled, THT, 2 A, 30 V

DATASHEET:[694103303002 Datasheet WR-DC DC Power Jack Right Angled THT](#)

Electrical Specifications

- **Rated Voltage:** 20 V DC (maximum)
- **Current Rating:** 3 A (maximum)

The barrel jack is used as the **DC power input connector** of the circuit. It allows an external adapter to safely supply power to the PCB. The built-in switching contact can automatically disconnect a backup battery or detect plug insertion when external power is connected. It acts as the main interface between the power supply and the electronic circuit.

2.USB PORT:

DESCRIPTION:USB 2.0 Receptacle Connector 4 Position Through Hole, Right Angle
USB-A-S-X-X-TH-R

DATASHEET:<https://www.snapeda.com/parts/USB-A-S-F-B-TH-R/Samtec/datasheet/>

Electrical Specifications

- **USB Standard:** USB 2.0
- **Rated Voltage:** 30 VAC / DC maximum
- **Current Rating:** Typically up to 1.5 A per USB 2.0 Type-A connector; actual circuit current depends on PCB design and protection components.
- **Data Transfer Rate:** Up to 480 Mbps (USB 2.0 High Speed)

The USB-A receptacle provides a standard interface for connecting USB devices such as keyboards, flash drives, sensors, or microcontrollers to the PCB. It carries both **5 V power** and **USB data signals (D+ and D-)** between the external device and the circuit. The connector serves as the communication and power entry/exit point for USB peripherals.

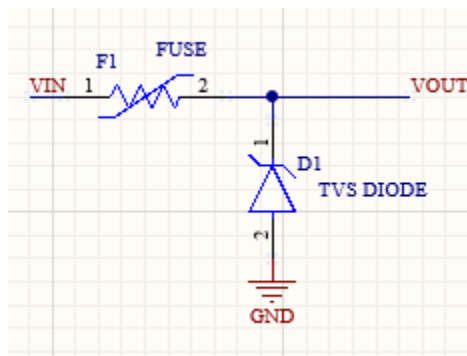
3.2 PIN POWER SUPPLY HEADER

DESCRIPTION: Terminal Block, 5MM Pitch, Horizontal, Pressure Clamp Glow Wire, Pcb, 2PINS, Gr

DATASHEET: [694103303002 Datasheet WR-DC DC Power Jack Right Angled THT](#)

The 2-pin header is used to connect an external power supply to the PCB. One pin carries the positive supply voltage (VCC) while the other provides the ground (GND) connection. It offers a simple and reliable method for supplying power during testing, development, or normal operation. The header acts as the entry point through which electrical power is delivered to the entire circuit.

PROTECTION CIRCUITRY:



1.FUSE:

DESCRIPTION: PTC RESET FUSE 33V 750MA 1812

DATASHEET: [1812L075-33DR-Littelfuse-datasheet-181956373.pdf](#)

Electrical Specifications

- **Rated Voltage (Max):** 33 V DC
- **Holding Current (Ihold):** 750 mA (0.75 A)
- **Trip Current (Itrip):** 1.5 A
- **Maximum Fault Current:** 20 A
- **Package Size:** 1812 (4.5 mm × 3.2 mm) SMD

The PTC resettable fuse protects the circuit from overcurrent and short-circuit conditions. Under normal operation it allows up to **750 mA** continuous current to flow. If excessive current occurs,

the fuse trips by increasing its resistance and limiting the fault current. After the fault is cleared, it automatically resets, providing reusable protection for the power input and connected components.

2.TVS DIODE:

DESCRIPTION:TVS DIODE 33V 53.3V SMB

DATASHEET:[SMB](#)

Electrical Specifications

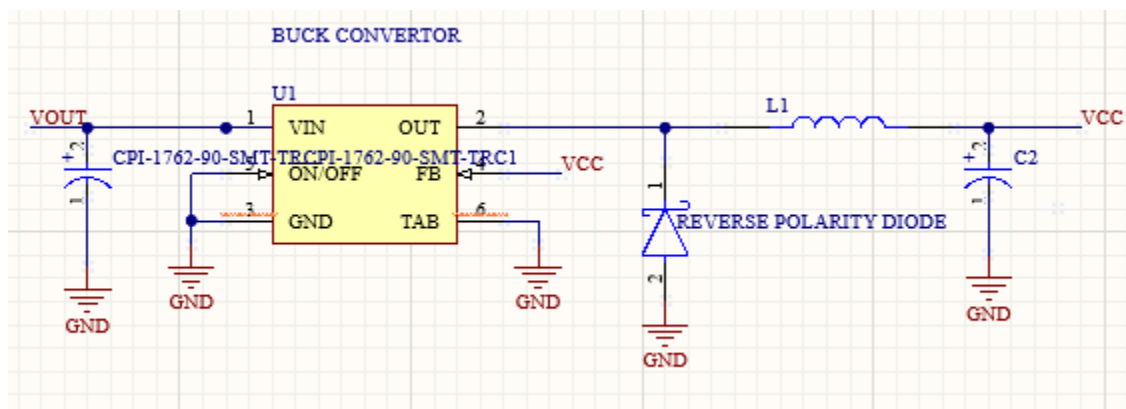
- **Working Standoff Voltage (VRWM):** 33 V
- **Breakdown Voltage (VBR):** Approximately 36.7 V – 40.6 V
- **Clamping Voltage (VC):** 53.3 V (maximum)
- **Peak Pulse Power:** Typically 600 W (10/1000 μ s waveform)
- **Peak Pulse Current (IPP):** Approximately 11.3 A
- **Package Type:** SMB (Surface Mount Package)

The 33 V TVS diode protects the circuit from sudden voltage surges and transient spikes on the power input line. Under normal conditions it remains inactive and does not affect circuit operation. During an overvoltage event, it conducts heavily and clamps the voltage to about 53.3 V, preventing damage to sensitive components. It is commonly placed across the power supply input as the primary surge protection device.

REFERENCE:<https://blog.mbedded.ninja/electronics/circuit-design/esd-protection/>

POWER SUPPLY UNIT:

The power supply circuit uses the LM2596-5.0 buck converter to step down 24 V DC to a regulated 5 V DC output. A reverse polarity diode protects the circuit from incorrect power connections. 680 μ F capacitors filter and stabilize the input voltage, while the 33 μ H inductor stores and transfers energy during switching operation. The 220 μ F output capacitor reduces ripple and ensures a stable 5 V supply for the load.



1. Buck converter:

datasheet: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.ti.com/lit/ds/symlink/lm2596.pdf>

Electrical Specifications

- **Input Voltage Range:** 4.5 V to 40 V DC
- **Output Voltage:** Fixed 5.0 V DC
- **Output Current:** Up to 3 A
- **Type:** Step-Down (Buck) Switching Regulator

The LM2596-5.0 converts the 24 V DC input into a regulated 5 V DC output. It uses high-frequency switching to achieve high efficiency with low power loss. The IC controls the switching action and maintains a stable 5 V output even when the input voltage or load current changes.

2. Reverse Polarity Protection Diode

datasheet: [SS34-E3-57T-Vishay-datasheet-181475108.pdf](#)

The diode protects the circuit if the power supply is connected with incorrect polarity. During correct operation it allows current to pass normally. If the input polarity is reversed, the diode blocks the current flow and prevents damage to the LM2596 and other components.

3. 680 μ F Input Capacitor

The 680 μ F capacitor at the input filters noise and voltage fluctuations from the 24 V supply. It acts as an energy reservoir and supplies transient current during switching operations. This helps maintain a stable input voltage to the buck converter.

5. Inductor

Electrical Value

- **Inductance:** 33 μ H

The inductor is the main energy storage element of the buck converter. It stores energy when the LM2596 switch is ON and releases it when the switch is OFF. This action smooths the current flow and helps reduce output voltage ripple.

6. 220 μ F Output Capacitor

The 220 μF capacitor filters the output of the buck converter and reduces voltage ripple. It stores charge and delivers current during sudden load demands. This helps maintain a stable and smooth 5 V output for the connected electronics.

CONTROL UNIT:

1. PIC16F877A Microcontroller

DESCRIPTION: Enhanced FLASH Microcontroller, 8K (x14-Bit words) FLASH, 368 Bytes RAM, 40-Pin PDIP, Standard VDD Range, Industrial Temperature

DATASHEET: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://ww1.microchip.com/downloads/en/devicedoc/39582b.pdf>

Electrical Specifications

- **Operating Voltage (VDD):** 4.0 V – 5.5 V
- **Operating Voltage (Typical):** 5 V
- **Maximum Clock Frequency:** 20 MHz
- **I/O Pins:** 33

The PIC16F877A acts as the main control unit of the system. It reads inputs from sensors and switches, processes the data according to the programmed logic, and controls outputs such as displays, relays, LEDs, or communication modules. All system operations are coordinated by this microcontroller.

2. 100 nF Decoupling Capacitor (Between VCC and GND)

Electrical Specifications

- **Capacitance:** 100 nF (0.1 μF)
- **Connection:** Between VDD and VSS pins

The 100 nF capacitor acts as a decoupling or bypass capacitor. It filters high-frequency noise and voltage spikes present on the power supply lines. During sudden switching inside the microcontroller, it provides instantaneous current and helps maintain a stable 5 V supply. This improves the reliability of the PIC16F877A.

3. 16 MHz Crystal Oscillator

DESCRIPTION:Crystal 16MHz ± 10 ppm 18pF SMD-4 3.2mm x 2.5mm

DATASHEET:[abm8](#)

Electrical Specifications

- **Frequency:** 16 MHz
- **Load capacitance:**18pF
- **Frequency Tolerance:** Typically ± 20 to ± 50 ppm

The 16 MHz crystal oscillator provides the clock signal required for the microcontroller operation. It determines the execution speed of instructions and timing functions. A stable crystal ensures accurate timing, communication, and overall system performance.

4. 22 pF Capacitors Connected to OSC1 and OSC2

Electrical Specifications

- **Capacitance:** 22 pF each
- **Connection:** One from OSC1 to GND and one from OSC2 to GND

The two 22 pF capacitors act as load capacitors for the 16 MHz crystal oscillator. They help the crystal start oscillating correctly and maintain frequency stability. These capacitors reduce noise and ensure reliable clock generation for the PIC16F877A.

5.MCLR Reset Circuit :

The MCLR reset circuit consists of a pull-up resistor, push-button switch, and capacitor connected to the MCLR pin of the PIC16F877A

1. Pull-up Resistor (Typically 10 k Ω)

The resistor keeps the MCLR pin at logic HIGH (5 V) during normal operation. Since MCLR is an active-low reset pin, keeping it high allows the microcontroller to run normally.

2. Push Button Reset Switch

When the push button is pressed, the MCLR pin is connected to GND, making it logic LOW. This resets the PIC16F877A and restarts program execution from the beginning.

3. Debouncing Capacitor (Typically 100 nF)

The capacitor filters noise and prevents false resets caused by switch bouncing or power supply disturbances. It also provides a short delay during power-up, ensuring the supply voltage becomes stable before the microcontroller starts operating.

EEPROM (Electrically Erasable Programmable Read-Only Memory)

DESCRIPTION:EEPROM 256Kbit, 400kHz, 2.5V, I2C Serial EEPROM, 8-Pin MSOP, Industrial Temperature

DATASHEET:<https://ww1.microchip.com/downloads/en/devicedoc/21203r.pdf>

Electrical Specifications

- **Operating Voltage:** 2.5 V – 5.5 V
- **Communication Interface:** I²C (SDA, SCL)
- **Operating Temperature:** -40°C to +85°C

The EEPROM is used to store data permanently even when power is removed from the circuit. It can store configuration settings, calibration values, user preferences, and system parameters. The PIC16F877A can read and write data to the EEPROM through the I²C interface whenever required. This ensures important information is retained after power loss.

3.3 V Voltage Regulator Circuit

Component: 3.3 V Linear Voltage Regulator (SOT-223-4 Package)

Datasheet: [UA78M 500mA, Positive-Voltage Linear Regulator datasheet \(Rev. V\)](#)

Electrical Specifications

- **Output Voltage:** 3.3 V Fixed
- **Maximum Output Current:** 500 mA
- **Regulator Type:** Linear Voltage Regulator
- **Package:** SOT-223-4
- **Input Voltage Range:** Typically 4.5 V to 15 V

The 3.3 V linear voltage regulator generates a regulated 3.3 V supply from the main power source. An input capacitor filters supply noise and stabilizes the input voltage, while an output capacitor reduces ripple and improves output stability. Together, these components provide a reliable 3.3 V power supply for low-voltage peripherals and communication modules used in the microcontroller board.

Wi-Fi Communication Module – ESP-01M (ESP8266)

DATASHEET: <https://www.alldatasheet.com/datasheet-pdf/view/1179098/ETC2/ESP-01.html>

Electrical Specifications

- **Module:** ESP-01M (ESP8266 Wi-Fi Module)
- **Operating Voltage:** 3.3 V
- **Operating Current:** 70–200 mA (typical during transmission)
- **Wi-Fi Standard:** IEEE 802.11 b/g/n
- **Frequency Band:** 2.4 GHz
- **Communication Interface:** UART (TX, RX)

The ESP-01M ESP8266 module provides Wi-Fi connectivity to the microcontroller board using the IEEE 802.11 b/g/n standard. It operates at 3.3 V and communicates with the PIC16F877A through UART TX and RX pins. Two resistors are used for logic-level interfacing and signal conditioning between the 5 V microcontroller and the 3.3 V Wi-Fi module. This enables wireless data transmission, remote monitoring, and IoT communication capabilities.

Temperature Monitoring Sensor – LM35DZ

Datasheet: [LP0003A LP \(O-PBCY-W3\) 4040001-2 and 4215214 \(Rev. E\)](#)

Electrical Specifications

- **Sensor Type:** Analog Temperature Sensor
- **Operating Voltage:** 4 V to 30 V
- **Temperature Range:** 0°C to +100°C

The LM35DZ is used to monitor the temperature of machines, motors, transformers, control panels, and industrial equipment. It detects abnormal temperature rises caused by overload, excessive friction, poor ventilation, or component failure. When the temperature exceeds a predefined limit, the microcontroller can generate an alarm, display a warning message, send a Wi-Fi notification, or shut down the equipment to prevent damage.

Opto-Isolation Circuit – PC817X1J000F Optocoupler

Datasheet: [pc817xi0000f e](#)

Electrical Specifications

- **Component Type:** Optocoupler / Photocoupler
- **Isolation Voltage:** 5 kVrms
- **Input LED Forward Voltage:** 1.2 V (typical)
- **Maximum Input LED Current:** 50 mA
- **Output Type:** Phototransistor
- **Maximum Collector-Emitter Voltage:** 80 V
- **Maximum Output Current:** 50 mA

The PC817 optocoupler provides **electrical isolation** between the microcontroller circuit and external circuits. It transfers signals using light, so there is no direct electrical connection between the input and output sides. This protects the PIC16F877A from high voltages, electrical

noise, ground loops, and transient disturbances. It is commonly used for industrial input monitoring, relay driving, and interfacing with external equipment.

OUTPUT SECTION:

1. Buzzer Driver Circuit

Electrical Specification

Operating Voltage - 3–5 V

Operating Current - 20–40 mA

The transistor acts as a switch to drive the buzzer because the PIC16F877A I/O pin cannot supply sufficient current directly. When the microcontroller output becomes HIGH, the transistor turns ON and energizes the buzzer. The buzzer provides an audible alarm for events such as high temperature, fault conditions, or system warnings. Red LED Indicator

LED INDICATOR

Electrical Specifications

Forward Voltage

1.8 V – 2.2 V

Operating Current

5–20 mA

The red LEDs are used as visual status indicators controlled by the PIC16F877A microcontroller. They provide immediate indication of system conditions such as power status, communication activity, alarm conditions, or operating modes. The LEDs improve system monitoring by allowing users to quickly identify the current state of the controller board.

Power Flow

24 V DC Input



PTC Fuse (750 mA)



TVS Diode Protection



Reverse Polarity Diode



LM2596 Buck Converter



5 V OUTPUT



PIC16F877A + LM35 + EEPROM+3.3V regulator+sensors



3.3 V OUTPUT



ESP-01M Wi-Fi Module+DISPLAY(if connected through I2C)

COMPONENT RATING:

Component	Operating Voltage	Current Rating
DC Barrel Jack (694106402002)	Up to 20 V DC (Rated)	3 A Max
USB Type-A Connector	5 V DC	Up to 500 mA (USB 2.0)
2-Pin Power Header	Depends on Supply	1–3A
PTC Resettable Fuse (1812L075)	33 V DC Max	750 mA Hold, 1.5 A Trip
TVS Diode (33 V SMB)	33 V Working Voltage	11.3 A Peak Pulse
Reverse Polarity Diode	24 V Input Line	Depends on Diode (Typically 1–3 A)
LM2596-5.0 Buck Converter	4.5–40 V Input	Up to 3 A Output
680 μ F Capacitors	≥ 35 V Rating (Typically)	Ripple Current Dependent
33 μ H Inductor	5 V Converter Circuit	Typically 3 A
220 μ F Capacitor	≥ 10 V Rating (Typically)	Ripple Current Dependent
PIC16F877A	4.0–5.5 V	10–20 mA Typical

Component	Operating Voltage	Current Rating
16 MHz Crystal	5 V Logic System	Few μ A
22 pF Capacitors	Oscillator Circuit	Negligible
100 nF Decoupling Capacitors	5 V / 3.3 V Rails	Negligible
EEPROM (24Cxx Series)	2.5–5.5 V	1–3 mA
3.3 V Linear Regulator	Input: 4.5–15 V	500 mA Max
ESP-01M (ESP8266)	3.0–3.6 V (3.3 V Typical)	80 mA Avg, 300 mA Peak
LM35DZ Temperature Sensor	4–30 V	60 μ A Typical
PC817 Optocoupler	LED Side: 1.2 V	5–20 mA Typical
MCLR Reset Switch Circuit	5 V	Very Low (<1 mA)
Red LED Indicators	1.8–2.2 V Forward Voltage	10–20 mA
Buzzer (5 V)	3–5 V	20–40 mA

Component	Operating Voltage	Current Rating
Buzzer Driver Transistor	Up to 45 V	Up to 100 mA Collector Current

Features

- 1. Industrial-Grade Power Protection**
 - Incorporates PTC fuse, TVS diode, and reverse polarity protection for robust operation in harsh environments.
- 2. Built-in Wi-Fi Connectivity**
 - ESP8266 module enables wireless monitoring and IoT communication.
- 3. Temperature Monitoring Capability**
 - Integrated LM35 sensor for real-time temperature sensing and machine health monitoring.
- 4. Optical Isolation Interface**
 - PC817 optocoupler provides electrical isolation between field signals and the microcontroller.
- 5. Non-Volatile Data Storage**
 - External EEPROM stores configuration and system data even during power loss.
- 6. Industrial IoT Ready Design**
 - Combines sensing, wireless communication, protection, and control functions on a single board.